CCIT Report #522 March 2005



# Bounds on Modified SPT Heuristics for Single Machine Flow-Time Scheduling with a Single Breakdown

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**Abstract**

Consider the problem of scheduling $n$ jobs with known processing time on a single machine. The machine undergoes a breakdown at a fixed point in time, known in advance, during the processing of the jobs. The job preempted due to the breakdown has to be restarted from the beginning. The objective function is minimizing the sum of completion times of the jobs. It was found that the problem is NP-hard and the SPT heuristic has a tight bound of 9/7. In this paper, we present a scheme of a series of heuristic algorithms as a function of the number of jobs scheduled before the breakdown. Every algorithm is characterized, for the worst case, by its bound and complexity. Thus, we provide the scheduler with a simple tool for determining his/her choice in the trade-off between accuracy of the solution (deviation from the optimum) and the computational effort.

**Index Terms**

Deterministic Scheduling, Minimum Flow-Time, Preemptive Repeat, Bounds, Heuristics

## I. Preliminaries

Consider the problem of scheduling $n$ jobs with known processing time on a single machine. The machine undergoes a breakdown at a fixed point in time, known in advance, during the processing of

This research was supported by the Fund for the Promotion of Research at the Technion.

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the jobs. The job preempted due to the breakdown has to be restarted from the beginning, namely, the processing time invested in it prior to the preemption is lost (i.e., "preemptive repeat"). The objective function is minimizing the flow time, or in other words, minimizing the sum of completion times of the jobs.

The problem is designated by $1/1brk/\sum C_i$, where $C_i$ denotes the completion time of job $i$ and $1brk$ stands for a single breakdown. In real life, we have situations where a machine is unavailable for a time interval that is known in advance; thus, the problem $1/1brk/\delta$ ($\delta$-an objective function) models realistic situations (for example, somebody who is going to leave for a vacation, or a machine due for periodic maintenance, etc.). Adiri et al. [Adiri, 1989] and Danusaputro and Lee [Danusaputro, 1992], proved that the problem is NP-hard. This creates the need for fast heuristics which construct schedules close to the optimal one. The quality of a heuristic algorithm is usually measured by two parameters: (i) The worst-case performance ratio. The heuristic algorithm A has an upper [lower] bound $b^A$ if, for any instance of the problem, the inequality $\frac{F(S^A)}{F(S^*)} < [>] b^A$ holds, where $F(S^E)$ is the flow time (sum of completion times) of schedule $S^E$, $E \in [A, *]$, $S^*$ denotes the optimal schedule. The bound is tight if there exists an instance such that either $\frac{F(S^A)}{F(S^*)} = b^A$ or at least $\frac{F(S^A)}{F(S^*)} \to b^A$. (ii) The complexity. The computational effort required, in the worst case, for constructing the schedule.

For the described $1/1brk/\sum C_i$ problem, Danusaputro and Lee [Danusaputro, 1992] proved that the SPT heuristic constructs, in $O(n \log n)$ steps, a schedule with a tight worst-case bound of 9/7, i.e., the maximum deviation from the optimum is 2/7 (28.6%).

In this paper, we analyze the SPT heuristic and suggest some possible improvements. Since the jobs are ordered according to the SPT rule that minimizes the sum of completion times, the only way to improve the result, namely, reducing the worst-case upper bound, is by interchanging jobs scheduled before the breakdown with jobs scheduled after it.

Let $|B^{SPT}|$ and $|A^{SPT}|$ denote the number of jobs scheduled, by the SPT rule, before and after the breakdown, respectively. $|B^{SPT}| + |A^{SPT}| = n$, where $n$ is the total number of jobs. Let $|B^{SPT}| - k$, $k \geq 0$, be the number of jobs scheduled before the breakdown by a modified SPT heuristic. In the sequel: (1) We analyze the case where $k = 0$, i.e., the number of jobs scheduled before the breakdown is the same in the modified and the original SPT heuristic. We distinguish between two cases here: (a) a single interchange, and (b) multiple interchanges. In (a), we start with the original SPT schedule and examine all possible feasible one-to-one interchanges of a job scheduled before the breakdown with one scheduled

after it and choose the one that yields the minimum flow time. Thus, the resulting schedule differs from the SPT by not more than one job. The complexity of this heuristic algorithm is $|A|\,|B|$, namely, $O(n^2)$. We designate this procedure by $k=0\ PSPT$. In (b), we check all feasible schedules for which the number of jobs scheduled before the breakdown equals that of the SPT schedule. Thus, SPT and $k=0\ PSPT$ are examined as well, and we choose the schedule that yields the minimum flow time. The number of schedules examined by this procedure is of order $\binom{n}{|B|}$, i.e., its complexity is not polynomial. We designate this heuristic algorithm by $k=0\ MSPT$. However, we demonstrate that both the $k=0\ PSPT$ and the $k=0\ MSPT$ have the same tight bound of $\frac{20}{17}$, i.e., the maximum deviation from the optimum is 17.6%, thus, an improvement of 35% over the SPT heuristic (from 28.6% to 17.6%). For further elaboration, see Example 1, where we exhibit the algorithms and show the differences among them, and the fact that they do not yield an optimal schedule.

**Example 1**: Consider an instance of the $1/1brk/\sum C_i$ problem with $n$ jobs. The breakdown time is $R+1$ and the repair time interval is $L$, $L'=L+1$. The processing times of jobs $t_j,\ 1\le j\le n$, are $t_1=t_2=\frac{R}{3}$; $t_3=t_4=\frac{R}{2}$; $t_{4+i}=R+1,\ i=1,\ldots,h$; $n=h+4,\ n\ge\lceil\frac{5R}{3}\rceil+L+6$ (see Figure 1). As expected, we have: $\sum C_i(OPT)\le\sum C_i(MSPT)<\sum C_i(PSPT)<\sum C_i(SPT)$.

(2) In the more general case of $k=m>0$, we examine schedules in which the number of jobs scheduled before the breakdown is greater than or equal to $|B^{SPT}|-k$, $k=1,2,\ldots,m,\ m<|B^{SPT}|-1$. We distinguish between two heuristic algorithms: $k=m\ PSPT$ and $k=m\ MSPT$. The $k=m\ PSPT$ starts with the SPT schedule and examines all feasible schedules where $x$, $1\le x\le m+1$, jobs before the breakdown are interchanged with $i$, $1\le i\le x$, jobs after it and chooses the one with minimum flow time. Thus, if the best schedule is found in stage $k=l\le m$, then the selected schedule is the best feasible obtained by replacing $l$ jobs before the breakdown with $x$, $x\in\{1,2,\ldots,l\}$ after it. We have that the $k=m\ PSPT$ heuristic chooses the schedule with minimum flow time out of those, where $x$, $x\in\{m+1,m,\ldots,1\}$ jobs before the breakdown are interchanged with $y$, $y\in\{1,2,\ldots,x\}$ jobs after it. Thus, the constructed schedule might differ from the original SPT by not more than $m+1$ of the jobs scheduled before the breakdown, while their number might be greater by not more than $m$.

The complexity of this algorithm is of order $O(n^{min[2(m+1),\frac{n}{2}]})$.

The $k=m\ MSPT$ chooses the best feasible schedule out of all those where the number of jobs scheduled before the breakdown is greater than or equal to $|B^{SPT}|-k$, $k=1,2,\ldots,m$, excluding schedules where the jobs scheduled prior the breakdown are a sub group of $B^{SPT}$, thus the number of

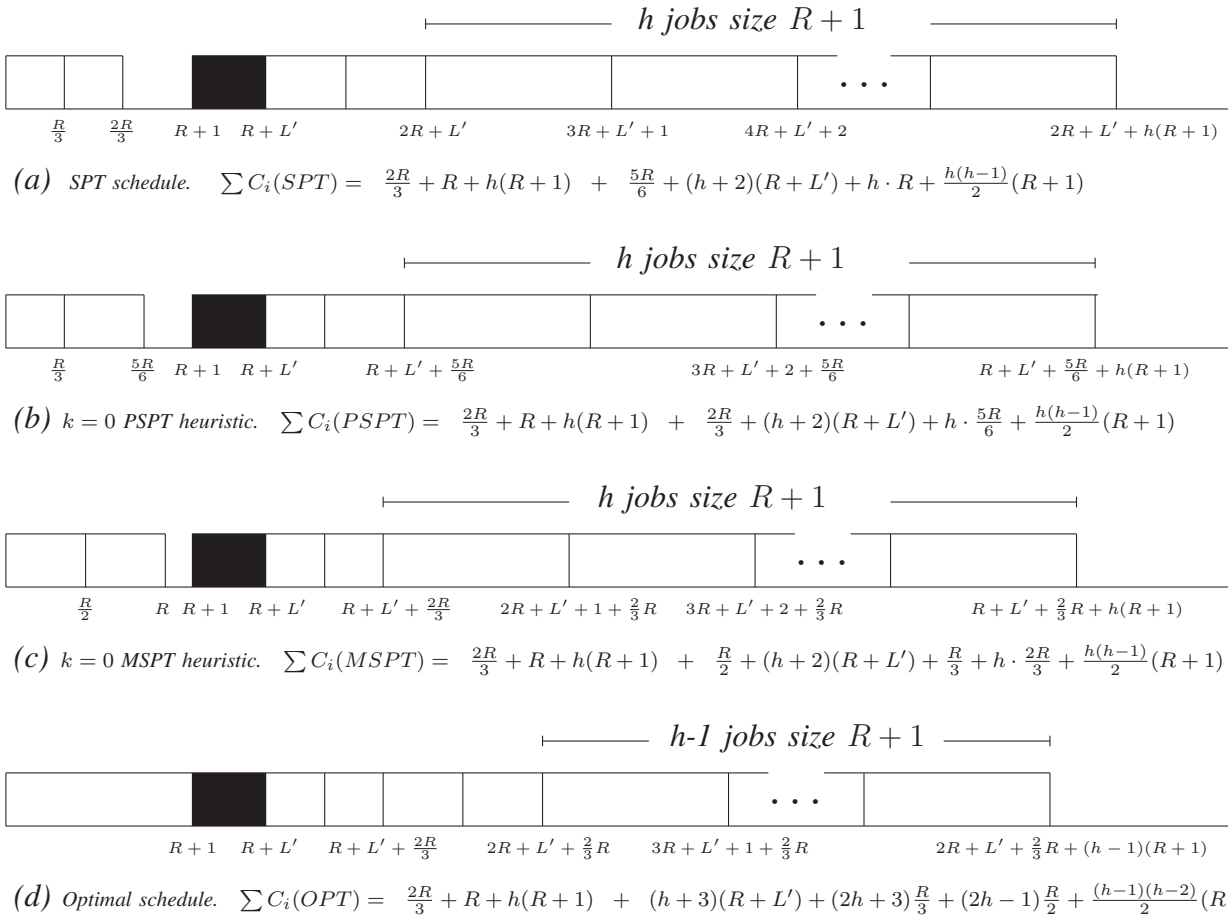



Fig. 1. Schedules for Example 1

schedules considered by $k = m > 0$ $MSPT$ is $\sum_{k=1}^{m} \binom{n}{B^{SPT}-k}$, i.e. the complexity of the algorithm is $O(n^{\frac{n}{2}})$. In agreement with our expectations the complexity of the *k=m PSPT* is not greater than that of the *k=m MSPT* heuristic. Moreover, the number of schedules examined by *k=m PSPT* never exceeds that of the *k=m MSPT* heuristic*.

We demonstrate that both $k = m$ $PSPT$ and $k = m$ $MSPT$, have the same tight bound of $1 + \frac{m+3}{2m^2+12m+17}$.

(3) Finally, we discuss the obtained results, reach some conclusions, and make some concluding remarks.

Although the $k = m \geq 0$ MSPT heuristic examines more schedules than the $k = m \geq 0$ PSPT, it does not yield a better bound. Thus, in view of the worst-case scenario, the PSPT with less computational effort guarantees the same result as the MSPT, i.e., the PSPT is preferable.

*For a given feasible $B^{SPT}$ the number of schedules checked by the *k=m MSPT* heuristic is $\sum_{l=1}^{|B^{SPT}|} \sum_{j=max[1,l-m]}^{l} \binom{|B^{SPT}|}{l} \binom{|A^{SPT}|}{j} + 1$ exchanging variables and the order of summation we have: $\sum_{k=1}^{m+1} \sum_{i=l}^{|B^{SPT}|} \binom{|B^{SPT}|}{i} \binom{|A^{SPT}|}{i-k+1} + 1 \geq \sum_{k=1}^{m+1} \sum_{i=k}^{m+1} \binom{|B^{SPT}|}{i} \binom{|A^{SPT}|}{i-k+1} + 1$ = *the number of schedules examined by k=m PSPT heuristic.*

The results are summarized in a table that shows (for the relevant values) the maximal deviation from the optimum as a function of $m$. Thus, we provide the scheduler with a simple tool for determining his choice in the trade-off between the computational effort and the deviation from the optimum.

## II. Mathematical models and their analysis

For relative simplicity of the analysis, explanations and the notations involved, we first focus on the simpler cases of the $k = 0$ $SPT$ heuristics, and eventually obtain a tight bound on the flow time. We then proceed to the more general cases of the $k = m \geq 0$ $SPT$. Since the analysis follows in part the same guidelines as for the $k = 0$ cases, some of the proofs are omitted. Finally, we obtain the tight bounds on the flow times as a function of $m$. In the mathematical analysis we use some ideas of [2].

### (1) $k = 0$ SPT heuristics

In these heuristics, we keep the same number of jobs scheduled before the breakdown as in the SPT schedule, namely, we may interchange the same number of jobs scheduled before the breakdown with those scheduled after it. We distinguish between two cases: (a) We start with an SPT schedule and allow a single one-to-one interchange, thus, the jobs scheduled before the breakdown according to this heuristic differ from their counterparts in the original SPT schedule by not more than one job. We examine the flow times obtained by the SPT schedule and by all feasible schedules resulting from a single interchange of any two jobs in it, and choose the one that yields the minimum flow time. Note that we allow interchange between any two jobs without the restriction that one be prior to the breakdown and the other after it, but the fact is that improvement of the flow time over the SPT schedule is only obtainable by this particular step. We denote this heuristic by $k = 0$ $PSPT$. (b) We examine all the feasible possibilities of choosing $|B^{SPT}|$ jobs out of the $n$ jobs and schedule them before the breakdown, and choose the one that yields the minimum flow time. Note that the schedules constructed by the $k = 0$ $PSPT$ are also examined now. We denote this heuristic by $k = 0$ $MSPT$.

*Notations*

Let us denote by $R$ the time when the breakdown occurs and by $L$ the repair time interval in which the machine is unavailable. Both $R$ and $L$ are known in advance, i.e., at time $0$. Sets $B^{SPT}$ and $A^{SPT}$ denote the jobs completed according to the SPT schedule before and after the breakdown, respectively.

We use the following notations:

$S^{SPT}$ - Schedule according to SPT.

$S^*$ - Optimal schedule.

$S^{0^1}$ - Schedule of $k = 0$ PSPT heuristic algorithm.

$S^0$ - Schedule of $k = 0$ MSPT heuristic algorithm.

$\delta^F$ - Time interval between the completion time of the last job before the breakdown and the time of the breakdown, for schedule $F$. ($F \in \{0'-$ for $k = 0\ PSPT$, $0-$ for $k = 0\ MSPT$, etc.$\}$).

$F(\sigma)$ - Flow time of schedule $\sigma$.

$F_V(\sigma)$ - Flow time of set $V$ in schedule $\sigma$.

$|E|$ - Number of jobs in set $E$, $E \in \{V, A, B, X, Y, \ldots\}$.

$X^*$ - First $|B^{SPT}|$ jobs in optimal schedule.

$Y^*$ - Remaining $|A^{SPT}|$ jobs in optimal schedule, namely, $|A^{SPT}| + |B^{SPT}| = |X^*| + |Y^*| = n$, where $n$ denotes the total number of jobs.

For better clarity, the $S^{SPT}, S^0$ and $S^*$ schedules are shown schematically in Figures 2a, 2b, 2c, respectively.

(a) $k = 0$ *PSPT Heuristic*

The main result for the $k = 0$ PSPT heuristic is stated in Theorem 1.

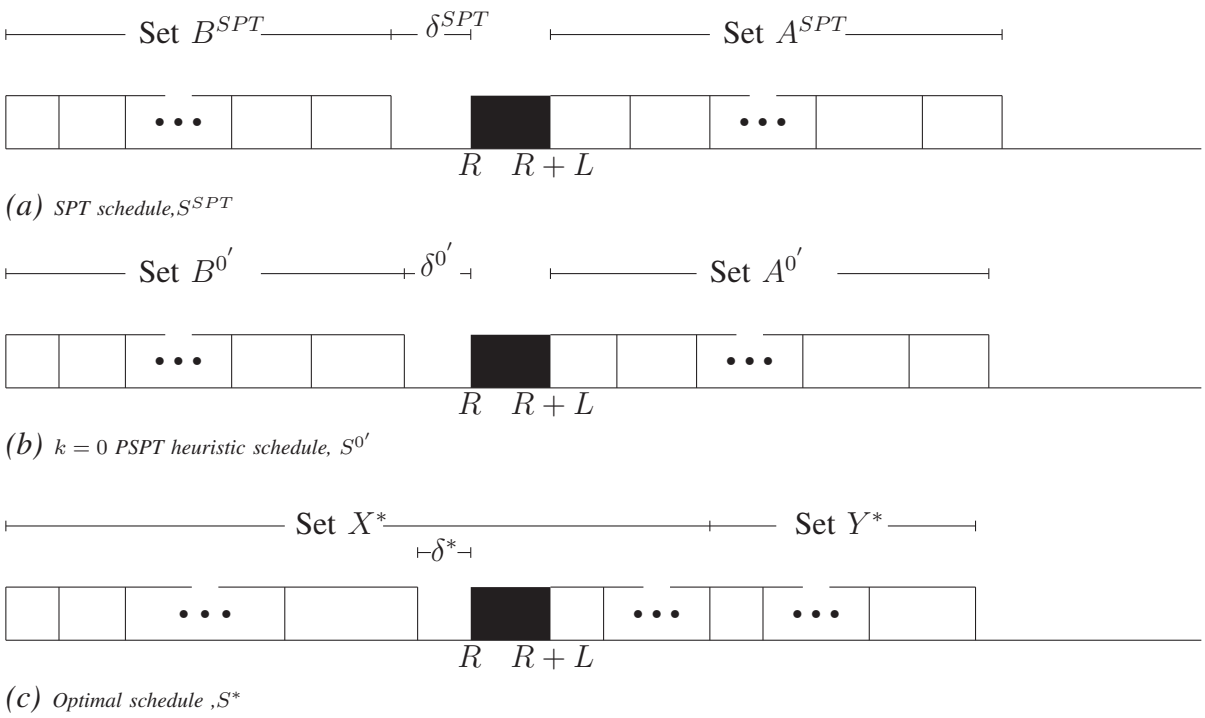
*Theorem 1:* The $k = 0$ PSPT heuristic has the tight lower bound of 20/17, i.e., the maximum deviation from the optimum is 18%.

The proof of the theorem is a direct consequence of four propositions formulated and demonstrated below.

*Proposition 1:*

$$|B^{SPT}| = |B^{0'}| \geq |B^*| \ . \tag{1}$$

*Proof:* In the SPT schedule $|B^{SPT}|$ jobs with the shortest processing time are scheduled before the breakdown, thus not more than $|B^{SPT}|$ jobs can be so scheduled, i.e., $|B^{SPT}| \geq |B^*|$. Interchange of one job before the breakdown with another after it does not change the number of jobs scheduled before the breakdown, i.e., $|B^{SPT}| = |B^{0'}|$. Note that $|B^{SPT}| = |B^*|$ implies that the $k = 0\ MSPT$ heuristic yields the optimal schedule, but does not necessarily imply that the $k = 0\ PSPT$ heuristic also yields the optimal schedule. □




*(a) SPT schedule,$S^{SPT}$*

*(b) $k=0$ PSPT heuristic schedule, $S^{0'}$*

*(c) Optimal schedule ,$S^*$*

Fig. 2. Definitions and notations

*Proposition 2:*

$$\delta^{0'} \leq \delta^{SPT} \geq \delta^* \tag{2}$$

*Proof:* $\delta^{0'} \leq \delta^{SPT}$ follows from the composition of the SPT schedule. Namely, this relation holds since in the SPT schedule the longest job before the breakdown is not bigger than the shortest one after it; thus, after an interchange of a job before the breakdown with one after it, the gap can only become smaller.

To prove that $\delta^{SPT} \geq \delta^*$ we argue as follows. Let us deduct $(\delta^{SPT} + L)$ from the completion time of all the jobs that belong to the set $A^{SPT}$ (those scheduled after the breakdown in the SPT schedule); and let us also deduct $(\delta^* + L)$ from the completion time of all the jobs that belong to the set $A^*$ (those scheduled after the breakdown in the optimal schedule). Following the deductions, the sums of completion times of the jobs in the SPT and the optimal schedules take the forms $F(S^{SPT}) - |A^{SPT}|(\delta^{SPT} + L)$ and $F(S^*) - |A^*|(\delta^* + L)$, respectively. The deductions eliminate the effect of the breakdown, and since the SPT schedule minimizes the sum of completion times of the jobs in the case of no breakdown, we arrive at:

$$F(S^{SPT}) - |A^{SPT}|(\delta^{SPT} + L) \leq F(S^*) - |A^*|(\delta^* + L) \tag{3}$$

Since $|B^{SPT}| \geq |B^*|$ or $|A^{SPT}| \leq |A^*|$, we have:

$$F(S^{SPT}) \leq F(S^*) - |A^*|(\delta^* - \delta^{SPT}) \tag{4}$$

or

$$|A^*|(\delta^* - \delta^{SPT}) \leq F(S^*) - F(S^{SPT}) \leq 0 \tag{5}$$

thus: $\delta^{SPT} \geq \delta^*$. □

*Proposition 3:* If $S^{0'} \neq S^*$, then

$$F(S^*) > \left\{ \frac{|Y^*|(|Y^*|+1)}{2} + 2 \right\} (\delta^{SPT} - \delta^*) \tag{6}$$

*Proof:* Since $S^{0'} \neq S^*$ there are at least two jobs that in the SPT schedule are scheduled before the breakdown and in the optimal schedule after it (otherwise we can achieve the optimal schedule by a single interchange in the SPT schedule). Thus the completion time of each of these two jobs is longer than $R > \delta^{SPT} \geq \delta^{SPT} - \delta^*$. Moreover, in addition to these two jobs there are $|A^{SPT}| = |Y^*|$ other jobs scheduled after the breakdown in the SPT schedule, thus their process times satisfy $p_{j \in A^{SPT}} > \delta^{SPT} \geq \delta^{SPT} - \delta^*$. Assuming that in the optimal schedule these jobs are scheduled before the breakdown, we have that the sum of their completion times is longer than $\frac{|Y^*|(|Y^*|+1)}{2}(\delta^{SPT} - \delta^*)$ (sum of completion times of $|A^{SPT}| = |Y^*|$ jobs with processing times $(\delta^{SPT} - \delta^*)$). Adding the completion times of the two jobs scheduled after the breakdown,we have Eq. 6. □

*Proposition 4:* If $S^{0'} \neq S^*$, then

$$F(S^{0'}) < F(S^*) + (|Y^*| - 2)\,(\delta^{SPT} - \delta^*) \tag{7}$$

*Proof:* Since $S^{0'} \neq S^*$ does not necessarily imply that $|B^{SPT}|$ is greater than $|B^*|$, it is possible that $|B^{SPT}| = |B^*|$ and $S^{0'}$ is not the optimal schedule ($F(S^*) < F(S^{0'})$); we will separate the proof into two cases, $|B^{SPT}| > |B^*|$ and $|B^{SPT}| = |B^*|$.

We start with the first case $|B^*| < |B^{SPT}|$. We note that $|B^{SPT}| \geq 2$, otherwise by a single one-to-one interchange we would have achieved the optimal schedule. Furthermore, since $|B^{SPT}| > |B^*|$ there is at least one job in set $X^*$ scheduled after the breakdown, i.e., $X^* \neq B^*$. The completion times of the last jobs in sets $B^{SPT}$ and $B^*$ are $R - \delta^{SPT}$ and $R - \delta^*$, respectively. Now we consider the last-but-one job

in $B^{SPT}$ and the last job in $X^*$. Clearly, the completion time of the first is shorter than $R-\delta^{SPT}$ while that of the second is longer than $R$, thus longer than $R-\delta^*$. Since the remaining jobs in set $B^{SPT}$ are the $(|B^{SPT}|-2)$ shortest ones, and are scheduled according to the SPT rule, their sum of completion times is not greater than that of their counterparts in the optimal schedule. Thus, we have

$$F_{B^{SPT}}(S^{SPT}) < F_{X^*}(S^*) - 2\left(\delta^{SPT} - \delta^*\right) \tag{8}$$

As a direct consequence of Proposition 2, we have that deducting $\delta^{SPT}$ from the completion time of every job in $A^{SPT}$ and $\delta^*$ from that of every job in $Y^*$, we arrive at:

$$F_{A^{SPT}}(S^{SPT}) - |A^{SPT}|\,\delta^{SPT} \leq F_{Y^*}(S^*) - |Y^*|\,\delta^* \tag{9}$$

Since $|A^{SPT}| = |Y^*|$, Eq. 9 takes the form:

$$F_{A^{SPT}}(S^{SPT}) \leq F_{Y^*}(S^*) - |Y^*|\left(\delta^* - \delta^{SPT}\right) \tag{10}$$

Eqs. 8 and 10 yield

$$F(S^{SPT}) < F(S^*) + (|Y^*| - 2)\,(\delta^{SPT} - \delta^*), \tag{11}$$

where $F(S^E) = F_{A^E}(S^E) + F_{B^E}(S^E), \quad E \in \{0', *, SPT\}$.
Since $F(S^{0'}) \leq F(S^{SPT})$, we have Eq.(7). □

This completes the proof for the first case where $|B^*| < |B^{SPT}|$. The proof for the second case where $|B^{0'}| = |B^{SPT}| = |B^*|$, is as follows.

Consider a schedule $S^{0^\dagger}$, where in the SPT schedule the last job scheduled before the breakdown is interchanged with its counterpart in the optimal schedule. This interchange is possible since $|B^{SPT}| = |B^*|$ and the first $|B^{SPT}|-1$ jobs in the SPT schedule are the smallest ones. Moreover, for $S^* \neq S^{SPT}$, the last job before the breakdown in the optimal schedule differs from the first $|B^{SPT}|$ jobs in the SPT schedule and is scheduled after the breakdown in the SPT schedule. Let us denote the last job scheduled before the breakdown in $S^{0^\dagger}$ (which is also the same according to $S^*$) by $J^\dagger$.

There are at least two jobs scheduled before the breakdown in the $S^{0^\dagger}$ schedule ($|B^{0^\dagger}| \geq 2$); otherwise $S^{0^\dagger}$ would be the optimal schedule. Their completion times are $R-\delta^\dagger$ and $R-\delta^\dagger-X_{J^\dagger}$, where $X_{J^\dagger}$ is the processing time of job $J^\dagger$. The completion times of the last two jobs scheduled before the breakdown according to $S^*$ are $R-\delta^*$ and $R-\delta^*-X_{J^\dagger}$. The completion times of the rest of the jobs in $B^{0^\dagger}$ are

not longer than those in $B^*$, since they are the $|B|-2$ shortest jobs. Hence we arrive at

$$F_{B^{0\dagger}}(S^{0\dagger}) \leq F_{B^*}(S^*) - 2\left(\delta^\dagger - \delta^*\right) \tag{12}$$

Deducting $\delta^\dagger + X_{J\dagger}$ from the completion times of the jobs in $A^\dagger$, and $\delta^* + X_{J\dagger}$ from those in $Y^*$, we have:

$$F_{A^{0\dagger}}(S^\dagger) - |A^\dagger|\left(\delta^\dagger + X_{J\dagger}\right) \leq F_{Y^*}(S^*) - |Y^*|\left(\delta^* + X_{J\dagger}\right) \tag{13}$$

Since $|A^\dagger| = |Y^*|$, we have

$$F_{A^{0\dagger}}(S^\dagger) \leq F_{Y^*}(S^*) - |Y^*|(\delta^* - \delta^\dagger) \tag{14}$$

Eqs. 12 and 14 yield:

$$F(S^\dagger) \leq F(S^*) + |Y^* - 2|(\delta^\dagger - \delta^*) \tag{15}$$

A straightforward consequence of Proposition 2 is that $\delta^\dagger < \delta^{SPT}$, and since $|Y^* - 2| \geq 0$, we have

$$F(S^\dagger)) < F(S^*) + |Y^* - 2|(\delta^{SPT} - \delta^*) \tag{16}$$

Since $F(S^{0'}) \leq F(S^\dagger)$ we arrive at Eq.(7).

This completes the proof for the case where $|B^{SPT}| = |B^*|$ and thus of Proposition 4. □

We now proceed to prove Theorem 1.

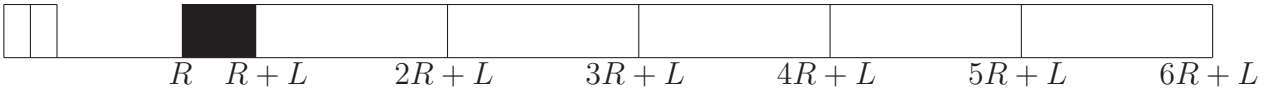
*Proof:* Combining Propositions 3 and 4, we have:

$$b^{0'} = \frac{F(S^{0'})}{F(S^*)} < 1 + \frac{(|Y^*| - 2)}{\left\{\frac{|Y^*|(|Y^*|+1)}{2} + 2\right\}} \tag{17}$$

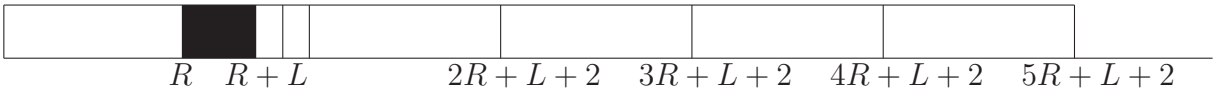
Applying standard techniques, we have that the bound is obtained for $|Y^*| = 5$, with the value of $\frac{20}{17}$. To complete the proof of Theorem 1, it remains to show that the bound is tight, in other words—to demonstrate that $b^{0'} \leq \frac{20}{17}$, i.e., to construct an instance that meets the bound. Consider an instance where $n = 7$ and the list of the processing times is $(1, 1, R, R, R, R, R)$, see Figure 3. Note that for this data ***SPT***, $k = 0$ ***PSPT*** and $k = 0$ ***MSPT*** construct the same schedule, $S^{SPT} = S^{0'} = S^0$.

$F(S^0) = 20R + 5L + 3$

$F(S^*) = 17R + 6L + 11$




*(a)* $k = 0$ *PSPT schedule,* $S^{0'}$




*(b) Optimal schedule,* $S^*$

Fig. 3. Demonstration that for the $k = 0$ *PSPT* heuristic the bound $\frac{20}{17}$ is tight.

For very large $R$ the ratio of $F(S^{0'})$ to $F(S^*)$ is $\frac{20}{17}$, thus the bound is tight. This completes the proof of Theorem 1. □

(b) $k = 0$ *MSPT heuristic*

This heuristic chooses the schedule with the minimum flow time out of all the feasible schedules where the number of jobs scheduled before the breakdown is the same as that in the *SPT* schedule. Clearly, the schedules constructed by *SPT* and $k = 0$ *PSPT* are included as well. Theorem 2 states that the worst-case bound for $k = 0$ *MSPT* is the same as that for $k = 0$ *PSPT* heuristic.

*Theorem 2:* The $k = 0$ *MSPT* heuristic has the same tight bound as the $k = 0$ *PSPT*, namely $\frac{20}{17}$.

*Proof:* Let us denote the bounds of $k = 0$ *PSPT* and $k = 0$ *MSPT* by $b^{0'}$ and $b^0$, respectively. Clearly the schedule constructed by $k = 0$ *MSPT* is at least as good as that constructed by $k = 0$ *PSPT*; thus $b^0 \leq b^{0'} = \frac{20}{17}$. We assume by contradiction that $b^0 < \frac{20}{17}$, i.e., for all feasible schedules where $|B^0| = |B^{SPT}|$ we have $\max\left(b^0 = \frac{F(S^0)}{F(S^*)}\right) < \frac{20}{17}$. However, for the job list as per Figure 3, we have $b^0 = b^{0'} = \frac{20}{17}$, which contradicts the assumption. Thus, $b^0 = \frac{20}{17}$ and the bound is tight. □

## (2) $k > 0$ SPT HEURISTICS

In this section, we deal with feasible schedules where the number of jobs scheduled before the breakdown is greater than or equal to $|B^{SPT}| - k$, $1 \leq k < |B^{SPT}| - 1$. Note that if $k \geq |B^{SPT}| - 1$, which means that it is also possible to schedule a single job before the breakdown (or none in the case of $k = |B^{SPT}|$), the optimal schedule can always be achieved since all feasible schedules are examined.

Two heuristic algorithms are studied, $k = m$ ***MSPT*** and $k = m$ ***PSPT***. The $k = m$ ***MSPT*** heuristic chooses the best feasible schedule, one with minimum flow time, out of all those in which the number of jobs scheduled before the breakdown is greater than or equal to $|B^{SPT}| - m$, $m < |B^{SPT}| - 1$. The $k = m$ ***PSPT*** heuristic chooses the best feasible schedule out of those where $x,\ x \in \{m+1, m, \ldots, 1\}$ jobs before the breakdown are interchanged with $y,\ y \in \{1, 2, \ldots, x\}$ jobs after it, and the remaining $|B^{SPT}| - x$ jobs before the breakdown and $|A^{SPT}| - y$ jobs after it remain the same as in the original ***SPT***.

Clearly, the feasible schedule chosen by the $k = m$ ***PSPT*** heuristic is one of those examined by the $k = m$ ***MSPT*** heuristic; thus the later schedule is at least as good as the former. However, we demonstrate that both schedules have the same tight bound of $1 + \frac{m+3}{2m^2+12m+17}$.

(a) $k = m$ *PSPT heuristic algorithm*

*Theorem 3:* The $k = m$ ***PSPT*** heuristic algorithm has a tight bound of $1+\frac{m+3}{2m^2+12m+17}$ for $m < |B^{SPT}|-1$ (for $m \geq (|B^{SPT}| - 1)$ the $k = m$ ***PSPT*** heuristic constructs an optimal schedule).

*Proof:* The proof follows similar guidelines to those used for Theorem 1, and is a straightforward result of Propositions 5–8.

Let us assume that the number of jobs scheduled before the breakdown according to the ***SPT*** schedule exceeds that of the optimal schedule by $m + r$ ($|B^{SPT}| - |B^*| = m + r$) where $r \geq 0$ (note that if $r = 0$, $k = m$ ***MSPT*** heuristic leads to the optimal schedule, but this is not necessarily the case for the $k = m$ ***PSPT*** heuristic).

*Proposition 5:* The optimal schedule contains at least $m + r + 1$ jobs scheduled after the breakdown while in the SPT schedule they are scheduled before it.

*Proof:* Since $|B^{SPT}| - |B^*| = m + r$, there are $m + r$ jobs scheduled after the breakdown in the optimal schedule, while in the SPT schedule they are scheduled before it. If those are the only jobs so scheduled, it means that the $|B^*|$ jobs scheduled before the breakdown in the optimal schedule are a subset of the $|B^{SPT}|$ jobs so scheduled in the SPT schedule, and in this case it is easy to verify that SPT is the optimal schedule. Hence, there must be at least one more job that according to the optimal schedule is scheduled after the breakdown, and in the SPT heuristic before it. □

Let $k'$ denotes a notation related to the $k = m$ ***PSPT*** heuristics.

*Proposition 6:*

$$\delta^{k'} \leq \delta^{SPT} \geq \delta^* \tag{18}$$

*Proof:* The proof is the same as that of Proposition 2. □

*Proposition 7:* If $S^{k'} \neq S^*$, then

$$F(S^*) > \left\{ \frac{|Y^*|(|Y^*|+1)}{2} + (m+r+1) \right\} (\delta^{SPT} - \delta^*) \tag{19}$$

*Proof:* According to Proposition 5 there are at least $m+r+1$ jobs that in the SPT schedule are scheduled before the breakdown, and in the optimal schedule after it. Thus, the completion time of each of these jobs is longer than $R > \delta^{SPT} \geq \delta^{SPT} - \delta^*$. Moreover, in addition to these $m+r+1$ jobs there are $|A^{SPT}| = |Y^*|$ other jobs scheduled after the breakdown in the SPT schedule, thus their process times satisfy $p_{j \in A^{SPT}} > \delta^{SPT} \geq \delta^{SPT} - \delta^*$. Assuming that in the optimal schedule these jobs are scheduled before the breakdown, we have that the sum of their completion times is not shorter than $\frac{|Y^*|(|Y^*|+1)}{2}(\delta^{SPT} - \delta^*)$ (sum of completion times of $|A^{SPT}| = |Y^*|$ jobs with processing times $(\delta^{SPT} - \delta^*)$). Adding the completion times of the $m+r+1$ jobs scheduled after the breakdown, we have Eq. 19. □

*Proposition 8:*

$$F(S^{k'}) < F(S^*) + (|Y^*| - (m+r+1))\,(\delta^{SPT} - \delta^*) \tag{20}$$

*Proof:* The proof of this proposition follows the guidelines of that of Proposition 4. We divide the proof into two cases, $|B^{SPT}| - |B^*| > m$ and $|B^{SPT}| - |B^*| \leq m$.

We start with the first case where $|B^{SPT}| - |B^*| = m + r, \quad r \geq 1$. The completion times of the last jobs before the breakdown in the SPT and in the optimal schedule are $R - \delta^{SPT}$ and $R - \delta^*$, respectively. In set $X^*$, there are $m+r$ jobs scheduled after the breakdown; thus their completion times are longer than $R$, hence also longer than $R - \delta^*$. The completion times of the $m+r$ jobs scheduled before the last job in set $B^{SPT}$ (jobs $J_{|B^{SPT}|-1}, J_{|B^{SPT}|-2} \cdots J_{|B^{SPT}|-(m+r+1)}$) are shorter than $R - \delta^{SPT}$. The completion times of the remaining jobs in set $B^{SPT}$ are not longer than those in set $B^*$, hence:

$$F_{B^{SPT}}(S^{SPT}) < F_{X^*}(S^*) - (m+r+1)\,(\delta^{SPT} - \delta^*) \tag{21}$$

As a direct consequence of Proposition 6, we have that deducting $\delta^{SPT}$ from the completion time of every job in $A^{SPT}$ and $\delta^*$ from that of every job in $Y^*$, we arrive at:

$$F_{A^{SPT}}(S^{SPT}) - |A^{SPT}|\,\delta^{SPT} \leq F_{Y^*}(S^*) - |Y^*|\,\delta^* \tag{22}$$

Since $|A^{SPT}| = |Y^*|$, Eq 22 takes the form:

$$F_{A^{SPT}}(S^{SPT}) \leq F_{Y^*}(S^*) - |Y^*| \left(\delta^* - \delta^{SPT}\right) \tag{23}$$

Eqs. 21 and 23 for $r \geq 1$ yield

$$F(S^{SPT}) < F(S^*) + (|Y^*| - 2)\,(\delta^{SPT} - \delta^*) \tag{24}$$

Since $F(S^{k'}) \leq F(S^{SPT})$, we have:

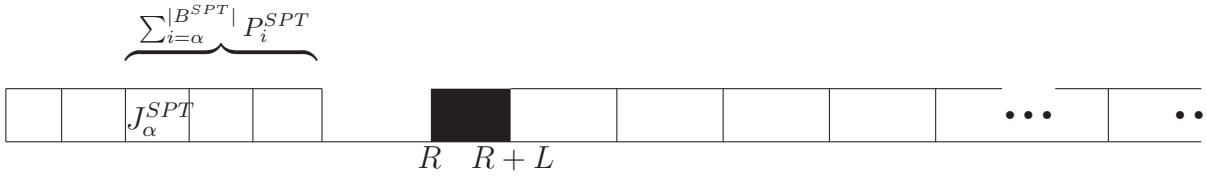
$$F(S^{k'}) \leq F(S^{SPT}) < F(S^*) + (|Y^*| - 2)\,(\delta^{SPT} - \delta^*) \tag{25}$$

This completes the proof for the case where $|B^{SPT}| - |B^*| > m$.

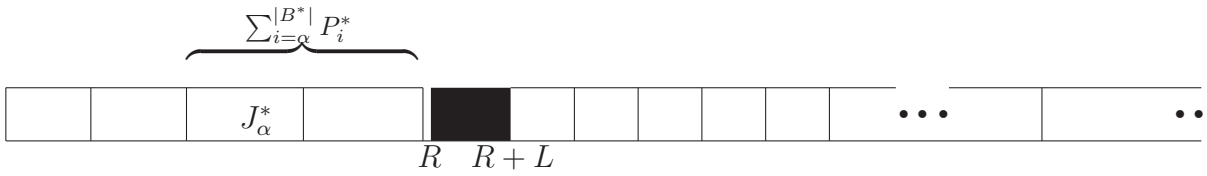
For the case where $|B^{SPT}| - |B^*| \leq m$, let us assume that $|B^{SPT}| - |B^*| = l, \quad 0 \leq l \leq m$. If $k = m$ ***PSPT*** heuristic does not lead to the optimal schedule there are at least $m - l + 2$ jobs scheduled before the breakdown according to the optimal schedule and after it according to the SPT heuristic, otherwise there are less than $m + 2$ jobs scheduled before the breakdown according to the SPT heuristic and not scheduled there according to the optimal schedule, hence the optimal schedule can be attained by the $k = m$ ***PSPT*** heuristic.

Let us denote by $J_i^*$ and $J_i^{SPT}$ the $i$-th job, scheduled from the start ($i-1$ jobs precede them) according to the optimal and SPT schedules respectively, let us also denote by $P_i^*$ and $P_i^{SPT}$ the processing times of jobs $J_i^*$ and $J_i^{SPT}$, respectively. Let us denote by $J_\alpha^*$ the last job scheduled before the breakdown according to the optimal schedule which satisfies the following relation: $\sum_{i=\alpha}^{|B^*|} P_i^* \geq \sum_{j=\alpha}^{|B^{SPT}|} P_j^{SPT}$, i.e. the last job before the breakdown such that the sum of its processing time and of those of the rest of the jobs following it and scheduled before the breakdown according to the optimal schedule, is greater than or equals that of the processing times of the counterpart job (the job with the same sequence number) and of those scheduled after it and before the breakdown according to the SPT schedule. For example if $|B^*| = |B^{SPT}|$ ($l = 0$) then $J_\alpha^*$ is simply $J_{|B^{SPT}|}^*$ since $\sum_{i=|B^{SPT}|}^{|B^*|} P_i^* = P_{|B^*|}^* \geq P_{|B^{SPT}|}^{SPT} = \sum_{j=|B^{SPT}|}^{|B^{SPT}|} P_j^{SPT}$ (equality only if SPT is the optimal schedule). Another example is given in Figure 4. Note that the sum $\sum_{j=\alpha}^{|B^{SPT}|} P_j$ contains $l$ more elements (jobs) than $\sum_{i=\alpha}^{|B^*|} P_i$, and also that there must be such a job, since according to Proposition 6 $\sum_{i=1}^{|B^*|} P_i \geq \sum_{j=1}^{|B^{SPT}|} P_j$.

We separate the proof for this case where $|B^{SPT}| - |B^*| = l, \quad 0 \leq l \leq m$ into two subcases, the first for $\alpha \leq |B^{SPT}| - (m+1)$, and the second for $\alpha > |B^{SPT}| - (m+1)$.




*(a) SPT schedule,* $S^{SPT}$




*(b) Optimal schedule,* $S^*$

Fig. 4. $J_\alpha^*$ is the last job scheduled before the breakdown according to both schedules which satisfies the following relation: $\sum_{i=\alpha}^{|B^*|} P_i^* \geq \sum_{j=\alpha}^{|B^{SPT}|} P_j^{SPT}$

We start with the first case. Let us compare the completion times of the jobs in the set $B^{SPT}$ with the ones in set $X^*$. The last $l$ jobs in the two sets are scheduled before and after the breakdown respectively, hence their completion times are smaller than $R-\delta^{SPT}$ for those in $B^{SPT}$, and greater than $R$ (which make them also greater than $R-\delta^*$) for those in $X^*$. Next let us compare the $m+2-l$ jobs which precede the last $l$ jobs in sets $B^{SPT}$ and $X^*$. Note that these jobs in set $X^*$ are the last jobs scheduled before the breakdown, which makes them the last $m+2-l$ in set $B^*$, while in the SPT schedule there are $l$ jobs scheduled after them and before the breakdown. The completion times of these jobs in set $B^{SPT}$ are $C_\beta^{SPT} = R-\delta^{SPT}-\sum_{i=\beta+1}^{|B^{SPT}|} P_i^{SPT}$ and those of the respective jobs in set $X^*$ are $C_\beta^* = R-\delta^*-\sum_{i=\beta+1}^{|B^*|} P_i^*$. Comparing them we have:

$$\begin{aligned} C_\beta^{SPT} &= R-\delta^{SPT}-\sum_{i=\beta+1}^{|B^{SPT}|} P_i^{SPT}-\delta^*-\sum_{i=\beta+1}^{|B^*|} P_i^*+\delta^*+\sum_{i=\beta+1}^{|B^*|} P_i^* = \\ & C_\beta^*-(\delta^{SPT}-\delta^*)-(\sum_{i=\beta+1}^{|B^{SPT}|} P_i^{SPT}-\sum_{i=\beta+1}^{|B^*|} P_i^*) \leq C_\beta^*-(\delta^{SPT}-\delta^*) \end{aligned}$$

where the last inequality holds for each job $J_\beta$ scheduled after $J_\alpha$, including the later. The completion times of the rest of the jobs in set $B^{SPT}$ (the first $|B^{SPT}|-(m+2)$ jobs) must be at least as small as the respective ones in set $X^*$, since they are the shortest. Summing it all up, we have:

$$F_{B^{SPT}}(S^{SPT}) < F_{X^*}(S^*)-(m+2)(\delta^{SPT}-\delta^*) \tag{26}$$

Deducting $\delta^{SPT}$ and $\delta^*$ from the completion times of all the jobs that belong to set $A^{SPT}$, and from those of all the jobs that belong to set $Y^*$, respectively, we have

$$F_{A^{kSPT}}(S^{SPT})) - |A^{SPT}|\,\delta^{SPT} \leq F_{Y^*}(S^*) - |Y^*|\,\delta^* \tag{27}$$

Since $|A^{SPT}| = |Y^*|$:

$$F_{A^{kSPT}}(S^{SPT})) \leq F_{Y^*}(S^*) - |Y^*|(\delta^* - \delta^{SPT}) \tag{28}$$

Eqs. 26 and 28 yield:

$$F(S^{SPT})) \leq F(S^*) + |Y^* - (m+2)|\,(\delta^{SPT} - \delta^*) \tag{29}$$

This completes the proof for the case $\alpha \leq |B^{SPT}| - (m+1)$.

We now turn to the case where $\alpha > |B^{SPT}| - (m+1)$.

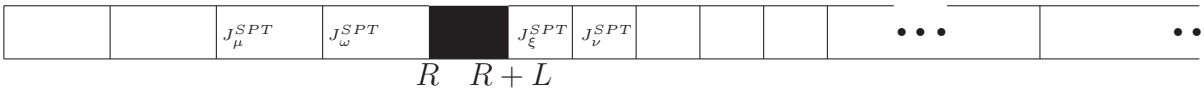
Let us look at a new schedule denoted by $S^{k^\dagger}$. According to it, the $m+1$ jobs scheduled right after the breakdown are those scheduled after the breakdown according to the optimal schedule (note that according to SPT these jobs are scheduled before the breakdown). The jobs scheduled before the breakdown according to $S^{k^\dagger}$ are the remaining $|B^{SPT}| - (m+1)$ jobs so scheduled according to SPT (at least one such job, since $|B^{SPT}| \geq m+2$), and after them but before the breakdown we schedule the $m+1-l$ jobs scheduled right before the breakdown according to the optimal schedule. An example of schedule $S^{k^\dagger}$ is given in Figure 5.

Note that $S^{k^\dagger}$ is simply interchanging $m+1-l$ jobs which according to SPT are scheduled somewhere after the breakdown, and according to the optimal schedule right before it, with $m+1$ jobs scheduled somewhere prior to the breakdown according to the SPT schedule and are the first $m+1$ scheduled right after it in the optimal schedule. Since $B^\dagger$ differs from $B^{SPT}$ by $m+1$, $S^{k^\dagger}$ is a schedule examined by the $k = m$ ***PSPT*** heuristic.

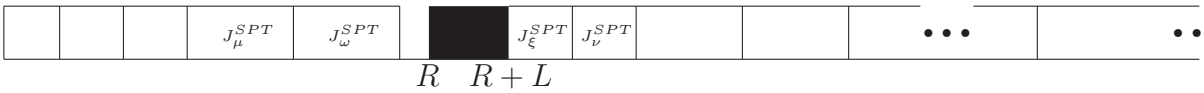
Note that $S^{k^\dagger}$ is feasible since before the breakdown it schedules $m+1-l$ jobs which are also so scheduled according to the optimal schedule, in addition to other $|B^*| - (m+1-l)$ jobs which have processing times not longer than their counterparts in the optimal schedule, hence the sum of processing times of the jobs scheduled prior to the breakdown in $S^{k^\dagger}$ is smaller than $R$. In order to prove Proposition 7 for this case we have to modify Proposition 6 and show that $\delta^\dagger \leq \delta^{SPT}$. The first $|B^{SPT}| - (m+1)$ jobs in $S^{k^\dagger}$ are also scheduled before the breakdown according to SPT, since they are not necessarily the first $|B^{SPT}| - (m+1)$ jobs in SPT their completion time (the sum of their processing times) is greater than




*(a) SPT schedule,*$S^{SPT}$




*(b) Optimal schedule,* $S^*$




*(a)* $k^{\dagger}$ *schedule,*$S^{k^{\dagger}}$

Fig. 5. An example of schedule $S^{k^{\dagger}}$

or equal to that of the first $|B^{SPT}| - (m+1)$ jobs in the SPT schedule, i.e.

$$\sum_{i=1}^{|B^{SPT}|-(m+1)} P_i^{\dagger} \geq \sum_{j=1}^{|B^{SPT}|-(m+1)} P_j^{SPT}$$

Since $\alpha > |B^{SPT}| - (m+1)$, the sum of the processing times of the rest of the $m+1-l$ jobs is not smaller than that of the last $m+1$ jobs scheduled before the breakdown according to the SPT schedule, i.e.

$$\sum_{i=|B^{SPT}|-(m+1)}^{|B^{\dagger}|} P_i \geq \sum_{j=|B^{SPT}|-(m+1)}^{|B^{SPT}|} P_j$$

Summing it all together, we have

$$\sum_{i=1}^{|B^{\dagger}|} P_i \geq \sum_{j=1}^{|B^{SPT}|} P_j$$

or

$$R - \sum_{i=1}^{|B^{\dagger}|} P_i \leq R - \sum_{j=1}^{|B^{SPT}|} P_j$$

which means that

$$\delta^{\dagger} \leq \delta^{SPT}$$

.

Now let us compare the completion times of the jobs according to schedules $S^{k^\dagger}$ and $S^*$. The completion time of job $J_\beta \in B^\dagger$ is $C^\dagger_\beta = R - \delta^\dagger - \sum_{i=\beta+1}^{|B^\dagger|} P^\dagger_i$ where the completion time of the corresponding job in the optimal schedule is $C^*_\beta = R - \delta^* - \sum_{i=\beta+1}^{|B^*|} P^*_i$. Comparing the two completion times for job $J_\beta$, which is one of the last $m+2-l$ jobs scheduled before the breakdown in the two schedules ($|B^\dagger|-(m+2-l) \leq \beta \leq |B^\dagger|$), we get $C^\dagger_\beta = C^*_\beta - (\delta^\dagger - \delta^*) - (\sum_{i=\beta+1}^{|B^\dagger|} P^\dagger_i - \sum_{i=\beta+1}^{|B^*|} P^*_i) = C^*_\beta - (\delta^\dagger - \delta^*)$, the last equality being due to the fact that the two sums are over the same set of jobs, hence are equal.

The processing time of each of the remaining jobs scheduled before the breakdown in $S^\dagger$ is at least as short as its counterpart in $S^*$, since they are the $|B^{SPT}| - (m+1)$ shortest jobs, not counting the $m+1$ jobs scheduled right after the breakdown in both schedules. Therefore the completion times of each of the $|B^\dagger| - (m+1-l)$ first jobs in $S^\dagger$ is not later than that of their counterparts in $S^*$. Note that if $S^\dagger$ is not the optimal schedule, there is at least one job whose processing time is shorter than the counterpart in $S^*$, hence its completion time is earlier than its counterpart in $S^*$. Comparing the sum of completion times of all jobs scheduled before the breakdown in both schedules, we have:

$$F_{B^\dagger}(S^\dagger) \leq F_{B^*}(S^*) - (m+2)(\delta^\dagger - \delta^*) \tag{30}$$

Deducting $\delta^\dagger + \sum_{i=|B^\dagger|-(m+1-l)}^{|B^\dagger|} P^\dagger_i - \sum_{i=|B^\dagger|}^{|B^\dagger|+l} P^*_i$ and $\delta^* + \sum_{i=|B^*|-(m+1-l)}^{|B^*|} P^*_i - \sum_{i=|B^*|}^{|B^*|+l} P^*_i$ from the completion times of the jobs in $A^\dagger$ and $A^*$ respectively results in:

$$F_{A^\dagger}(S^\dagger) - |A^\dagger|(\delta^\dagger + \sum_{i=|B^\dagger|-(m+1-l)}^{|B^\dagger|} P^\dagger_i - \sum_{i=|B^\dagger|}^{|B^\dagger|+l} P^*_i) \leq F_{A^*}(S^*) - |A^\dagger|(\delta^* + \sum_{i=|B^*|-(m+1-l)}^{|B^*|} P^*_i - \sum_{i=|B^*|}^{|B^*|+l} P^*_i) \tag{31}$$

The last inequality exists since by eliminating the gap between the last job scheduled before the selected jobs and the first job scheduled after them, and by ignoring the set of selected jobs which are scheduled in the same position, we are actually comparing the flow times of the list of the remaining jobs with no breakdown, for which SPT is optimal.

But since $|A^\dagger| = |A^*|$, $\sum_{i=|B^\dagger|-(m+1-l)}^{|B^\dagger|} P^\dagger_i = \sum_{i=|B^*|-(m+1-l)}^{|B^*|} P^*_i$ and $\sum_{i=|B^\dagger|}^{|B^\dagger|+l} P^*_i = \sum_{i=|B^*|}^{|B^*|+l} P^*_i$ according to the construction of $S^\dagger$, we have

$$F_{A^\dagger}(S^\dagger) \leq F_{A^*}(S^*) + |A^\dagger|(\delta^\dagger - \delta^* +) \tag{32}$$

This concludes the proof of the second part and thus of Proposition 8. □

Now we proceed to prove Theorem 3.

*Proof:* Combining Propositions 7 and 8, we have

$$\frac{F(S^{k'})}{F(S^*)} < 1 + \frac{(|Y^*| - (m+r+1))}{\left\{\frac{|Y^*|(|Y^*|+1)}{2} + (m+r+1)\right\}} \tag{33}$$

The maximum value is obtained for $r = 1$, thus the upper bound is:

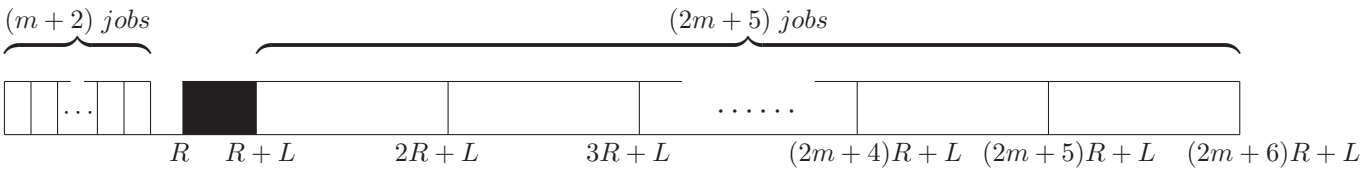
$$\frac{F(S^{k'})}{F(S^*)} < 1 + \frac{(|Y^*| - (m+2))}{\left\{\frac{|Y^*|(|Y^*|+1)}{2} + (m+2)\right\}} \tag{34}$$

In order to find the $|Y^*|$ associated with the worst error, we use standard techniques and after non-trivial mathematical manipulations arrive at $|Y^*| = 2m + 5$. Substitution of this value in Eq. 34 yields a bound as per Theorem 3, namely, $\frac{F(S^k)}{F(S^*)} < 1 + \frac{m+3}{2m^2+12m+17}$. Note that this result holds only for $m < |B^{SPT}| - 1$; we use this assumption to advantage in the proof of Proposition 5. Obviously, if we let $m$ equal $|B^{SPT}|$ or $|B^{SPT}| - 1$, we end up examining all possible schedules which lead to the optimal schedule.

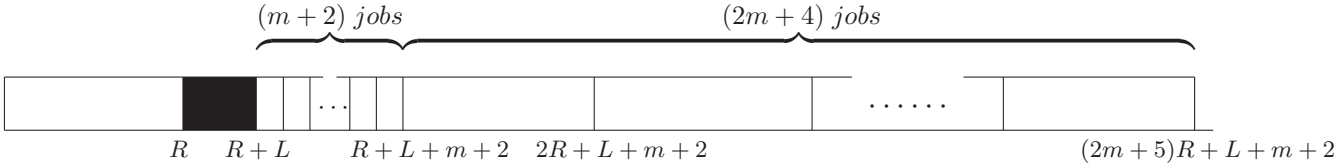
To complete the proof of Theorem 3, it remains to demonstrate that the bound is tight, i.e., to construct an instance of the $1/1brk/\sum C_i$ problem where $\frac{F(S^k)}{F(S^*)}$ meets the bound.

Consider $(3m + 7)$ jobs with the list of processing times $(\overbrace{1, 1, \ldots, 1}^{m+2}, \overbrace{R, R, \ldots, R}^{2m+5})$. Figures 6a and 6b show the schedules constructed by the $k = m$ ***PSPT*** heuristic and the optimal one, respectively.

Let us denote by $S^{SPT}$, $S^{kPSPT}$ and $S^{kMSPT}$ the schedules constructed by the SPT, $k = m$ ***PSPT*** and $k = m$ ***MSPT*** heuristics, respectively.




*(a) The schedule constructed by the $k = m$ PSPT heuristic, $S^{kPSPT}$*




*(a) The optimal schedule, $S^*$*

Fig. 6. Demonstration that for the $k = m$ *PSPT* heuristic the bound is tight.

Note that the schedule as per Figure 6a is the one constructed by all three heuristics (SPT, $k = m$ ***PSPT*** and $k = m$ ***MSPT***), since for this data $S^{SPT} = S^{kPSPT} = S^{kMSPT}$.

$$F(S^{kPSPT}) = \sum_{i=1}^{m+2} i + \sum_{i=1}^{2m+5} (R + L + iR) = \frac{(m+3)(m+2)}{2} + (2m+5)(R+L) + (m+3)(2m+5)R$$
$$\approx (m+4)(2m+5)R + O(R)$$

$$F(S^*) = R + \sum_{i=1}^{m+2}(R + L + i) + \sum_{i=1}^{2m+4} (R + L + m + 2 + iR) =$$
$$R + (m+2)(R+L) + \frac{(m+2)(m+3)}{2} + (2m+4)(R+L+m+1) + \frac{(2m+5)(2m+4)R}{2}$$
$$\approx (2m^2 + 12m + 17)R + O(R)$$

We have,

$$\frac{F(S^{k'})}{F(S^*)} = \frac{F(S^{kPSPT})}{F(S^*)} = 1 + \frac{m+3}{2m^2 + 12m + 17} \tag{35}$$

thus the bound is tight. This completes the proof of Theorem 3. □

(b) $k = m$ ***MSPT heuristic algorithm***

Although the $k = m$ ***MSPT*** heuristic, in its search for the schedule with minimum flow time, examines a larger number of schedules than the $k = m$ ***PSPT***, the bound on the flow time of the chosen schedule is the same as that of the latter.

Clearly, all the schedules examined by the $k = m$ ***PSPT*** heuristic are examined by the $k = m$ ***MSPT*** heuristic as well, but the later examines schedules not examined by the former.

Let us denote the in bounds by $b^{kPSPT}$ and $b^{kMSPT}$, respectively.

*Theorem 4:* The $k = m$ ***MSPT*** heuristic has the same tight bound as the $k = m$ ***PSPT*** heuristic, namely $b^{kMSPT} = b^{kPSPT} = 1 + \frac{m+3}{2m^2+12m+17}$.

*Proof:* Obviously, the schedule constructed by the $k = m$ ***PSPT*** is not better than the one constructed by the $k = m$ ***MSPT***; thus, $b^{kMSPT} \leq b^{kPSPT} = 1 + \frac{m+3}{2m^2+12m+17}$.

Let us assume by contradiction that $b^{kMSPT} < b^{kPSPT}$, i.e., for all feasible schedules where the number of jobs scheduled before the breakdown satisfies $\{|B| = |B^{SPT}| - k,\ k = 0, 1, \ldots, m\}$ we have, $\max\left(b^{kMSPT} = \frac{F(S^{kMSPT})}{F(S^*}\right) < 1 + \frac{m+3}{2m^2+12m+17}$. However, for the job list as per Figure 6, we have $b^{kMSPT} = b^{kPSPT}$. Contradiction. Thus, $b^{kMSPT} = 1 + \frac{m+3}{2m^2+12m+17}$ and the bound is tight. □

## III. CONCLUDING REMARKS

(i) In this paper, we analyze heuristic algorithms for solving the NP-hard problem $1/1brk/\sum C_i$. We present a series of approximate algorithms as a function of $k = m$. The larger the $m$, the closer the constructed schedule to the optimal one. However, as $m$ increases, so does the complexity of the algorithm and thus the computational effort, i.e, it takes more time to reach a solution. Thus, the scheduler has to make his choice in the trade-off between the accuracy of the solution and its cost in terms of the computational effort.

(ii) The concluding remarks refer to the $k = m \geq 0$ PSPT heuristics, not to the MSPT ones, since both have the same bounds and the computational effort of the former is lower, i.e., the $k = m \geq 0$ PSPT heuristics are preferable.

(iii) Table 1 below displays the worst-case bound (on $\frac{F(S^{kPSPT})}{F(S^*)}$) as a function of $m$. It starts with the SPT worst-case bound, proceeds to the $k = 0$ PSPT bound and continues with $k = m$ PSPT, $m = 1, 2, \ldots, |B^{SPT}| - 1$.

Table 1: Bounds and complexities for the $k = m$ PSPT heuristics

| $m$ | SPT | 0 | 1 | 2 | 3 | 4 | 5 | $\cdots$ | 9 | $\cdots$ |
|---|---|---|---|---|---|---|---|---|---|---|
| Bound | 1.29 | 1.176 | 1.13 | 1.10 | 1.08 | 1.07 | 1.06 | | 1.04 | |
| Complexity | $n \log n$ | $n^2$ | $n^4$ | $n^6$ | $n^8$ | $n^{10}$ | $n^{12}$ | | $n^{20}$ | |

Examining Table 1, we have that for the $1/1brk/\sum C_i$ problem, the maximum deviation from the optimum of the SPT heuristic is 29% and its complexity $O(n \log n)$; while that of the $k = 0$ PSPT is 17.6% and its complexity $O(n^2)$; and that of $m = 2$ is 10% and its complexity $O(n^6)$. From the worst-case point of view, for a $k = m \geq 3$ PSPT heuristic, the computational complexity is not smaller than $O(n^8)$, and is thus, in most cases, impractical. However, the $k = m \leq 2$ PSPT heuristics guarantee a solution within 10% from the optimum, and its computational effort does not exceed $O(n^6)$.

(iv) The $k = 0$ PSPT heuristic is given special detailed consideration for two reasons: (a) It is the most practical case, since its complexity is only $O(n^2)$, and it guarantees a solution within 17.6% of the optimum; (b) The relative simplicity of the notations and explanations, and the clarity of the mathematical analysis favor it as starting point.

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